

Please refer to **Appendix A** pp **A6-A9** of the recommended text book.

UNIT 1.2

THE ABSOLUTE VALUE

Definition of absolute value. please see the definition on page A6

Examples: Express the following expressions without the absolute value symbol.

(1) $|| - 2| - | - 3||$

(2) $|3x - 2|$

(3) $|x - 2|$ if $x < 2$

Geometric interpretation (meaning) of the absolute value.

Absolute values as functions.

- Discuss $f(x) = |x|$ as a function.

- Use the definition of the absolute value to write the following functions as piecewise defined functions and sketch their graphs

(1) $f(x) = |x - 1|$

(2) $f(x) = |x^2 - 1|$

Properties of absolute values. Please refer to page A7 (These are crucial for solving equations and inequalities involving absolute values).

Examples: using properties of absolute value to solve equations and inequalities involving absolute values.

(1) Solve $|1 - x| = 0$.

(2) Solve $|2x - 3| = 2$.

(3) Solve $|3x - 4| \geq 2$.

(4) Solve $|5x - 2| \leq 1$.

(5) Solve $|x^2 - 9| = 1$.

(6) Solve $|1 - x| < 0$.

(7) Solve $|-x| < 0$.

(8) Solve $\left|\frac{x}{x-2}\right| > 0$.

(9) Solve $\frac{x}{|x-2|} \geq 0$.

(10) Solve $\frac{|x+1|}{|x-2|} > 0$.